	$W \mapsto X^n$	$X^n \mapsto Y^n$	$Y^n \mapsto \hat{W}$	
	$\Omega \to \mathcal{X}^n$	$\mathcal{X} \to \mathcal{Y}$	$\mathcal{Y} \to \Omega$	
Message	Encoder	Channel	Decoder	Estimator
$W \in \Omega$	$X \in \mathcal{X}$	$P_{Y\|X}$	$Y^n \in \mathcal{Y} = \{0,1\}^n$	$\hat{W}$
	Alphabet			
L bits	n bits			

$M = 2^L$ cases

2 bits, 4 cases

$W = \gamma$, $M = 4$, $L = 2$

$\Omega = \{\alpha, \beta, \gamma, \theta\}$

P 1/2 1/4 1/8 1/8

$\xrightarrow[\text{Encoding}]{\text{Huffman}}$

$X^n = 001$, $n = 3$

$\mathcal{X} = \{1, 01, 001, 000\}$

$\to 100, 010, 001, 000$

$\xrightarrow[p=\frac{1}{16}]{\text{BSC}}$ $Y^n = 000$

$\text{Rate}(\gamma) = \frac{1}{n}\log M = \frac{2}{3}$ $\qquad C = 1 - H(p) = \frac{3}{4}$

$R < C$, so

[例]

Noiseless Binary Channel $C = 1$

$0 \to 0$, $1 \to 1$ $\quad \max_{p(x)} H(Y) - H(Y|X)$, $H(Y|X) = 0$, $p(x)$ 等概率时, $p(y)$ 等概率

BSC (Symmetric) $C = 1 - H(p)$

$0 \xrightarrow{1-p} 0$, $0 \xrightarrow{p} 1$, $1 \xrightarrow{1-p} 1$, $1 \xrightarrow{p} 0$ $\quad \max_{p(x)} H(Y) - H(Y|X)$, $H(Y|X) = H(p)$, $p(x)$ 等概率时, $p(y)$ 等概率

BEC (Erasure) $C = 1 - \alpha$

$0 \xrightarrow{1-\alpha} 0$, $0 \xrightarrow{\alpha} e$, $1 \xrightarrow{\alpha} e$, $1 \xrightarrow{1-\alpha} 1$ $\quad \max_{p(x)} H(Y) - H(Y|X)$, $H(Y|X) = H(\alpha)$

但做不到 $p(y)$ 等概率

$$H(Y) = H((1-\alpha)p, (1-\alpha)(1-p), \alpha)$$

$$= H(\alpha) + H((1-\alpha)p, (1-\alpha)(1-p) \mid \alpha) = H(\alpha) + (1-\alpha)H(p)$$

$0 \to 1$, $0 \to 2$, $1 \to 3$, $1 \to 4$ $\quad C = 1$

$\max_{p(x)} H(X) - H(X|Y)$, $H(X|Y) = 0$

对称信道：转移矩阵 P 各行各列为不同排列

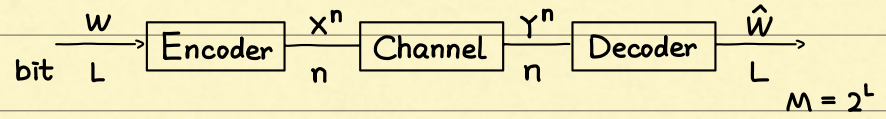
弱对称信道：各行不同排列，列和相同

则 $C = \log|Y| - H(\text{行向量})$

[例] $P = \begin{vmatrix} 1/3 & 1/2 & 1/6 \\ 1/6 & 1/3 & 1/2 \\ 1/2 & 1/6 & 1/3 \end{vmatrix}$ $\quad P = \begin{vmatrix} 1/3 & 1/2 & 1/6 \\ 1/3 & 1/6 & 1/2 \end{vmatrix}$

$C = 1 - H(1/3, 1/2, 1/6)$

$C = \log 3 - H(1/3, 1/2, 1/6)$



信源 w 共 $M = 2^L$ 种情况

$R = \frac{L}{n} = \frac{1}{n}\log M$

错误概率 $P_e = P(w \neq \hat{w})$

[例] $L = 1$，$n = 3$，$R = 1/3$，W：0 或 1，X^n：000 或 111

信道每位有 p 出错 Y^n 共 8 种情况，$\hat{w}$ 取 Y 中 0 多或 1 多

则 $P_e = 3p^2(1-p) + p^3$

Shannon 2nd Thm.

$R < C$，存在编码使 $P_e \to 0$